Kinetic energy: In physics, the kinetic energy of an object is the energy which it possesses due to its motion. It is defined as the work needed to accelerate a body of a given mass from rest to its stated velocity. Having gained this energy during its acceleration, the body maintains this kinetic energy unless its speed changes. The kinetic energy of an object of mass *m* traveling at a speed *v* is $K.E. = ½\, mv^2$.

Average kinetic energy of a vibrating particle:

The average kinetic energy of a vibrating particle is the sum of the kinetic energies over a complete time period divided by the time period.

The displacement and velocity of a vibrating particle is given by

$$x = a \sin(\omega t + \alpha)$$

$$v = dx/dt = a\omega \cos(\omega t + \alpha)$$

If m is the mass of the vibrating particle, the kinetic energy at any instant

$$= \frac{1}{2} mv^2 = \frac{1}{2} \text{m } a^2\, \omega^2 \cos^2(\omega t + \alpha) \qquad \textbf{Or,}\ \boldsymbol{K = \frac{1}{2}K\,(a^2 - x^2)}$$

The average kinetic energy of the particle in one complete vibration

$$= \frac{1}{T}\int_0^T \frac{1}{2} \text{m } a^2\, \omega^2 \cos^2(\omega t + \alpha)dt$$

$$= \frac{1}{T}\frac{ma^2\omega^2}{4}\int_0^T 2\cos^2(\omega t + \alpha)dt$$

$$= \frac{ma^2\omega^2}{4T}\int_0^T [1 + \cos 2(\omega t + \alpha)]dt$$

$$= \frac{ma^2\omega^2}{4T}[\int_0^T dt + \int_0^T \cos 2(\omega t + \alpha)dt]$$

$$But \int_0^T \cos 2(\omega t + \alpha)dt] = 0$$

Therefore average kinetic energy

$$= \frac{ma^2\omega^2}{4T}\text{T} + 0$$

$$= \frac{ma^2\omega^2}{4} = \frac{ma^2(4\pi^2n^2)}{4}$$

$$= \pi^2 \text{m } a^2 n^2$$

Potential energy: Potential energy is energy that is stored within an object. The potential energy of the particle is the amount of work done to move the particle by a distance against the restoring force. Let a particle of mass *m* executes simple harmonic motion. Let any instant of time *t* its displacement is *y*. Then the potential energy of the particle will be

$$P.E. = \frac{1}{2}kx^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.. (3)$$

Average potential energy of a vibrating particle:

The average potential energy of a vibrating particle is the sum of the potential energies over a complete time period divided by the time period.

The displacement and velocity of a vibrating particle is given by

x = a sin (ωt + α)

v = dx/dt = aω cos (ωt + α)

If m is the mass of the vibrating particle, the potential energy at any instant

$$= \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2a^2sin^2(\omega t + \alpha) \qquad Since\ \omega^2 = \frac{k}{m}$$

The average potential energy of the particle in one complete vibration

$$= \frac{1}{T}\int_0^T \frac{1}{2}m\omega^2a^2sin^2(\omega t + \alpha)\,dt$$

$$= \frac{1}{T}\frac{m\omega^2a^2}{4}\int_0^T 2sin^2(\omega t + \alpha)\,dt$$

$$= \frac{m\omega^2a^2}{4T}\int_0^T [1 - \cos 2\,(\omega t + \alpha)]\,dt$$

$$= \frac{m\omega^2a^2}{4T}\left[\int_0^T dt - \int_0^T \cos 2\,(\omega t + \alpha)\,dt\right]$$

But $\int_0^T \cos 2\,(\omega t + \alpha)\,dt = 0$

Therefore the average potential energy is $= \frac{m\omega^2a^2}{4T}T$

$$= \frac{m(2\pi n)^2a^2}{4}$$

$$= \pi^2ma^2n^2$$

Where m is the mass of the vibrating particle, *a* is the amplitude of vibration and n is the frequency of vibration. Also, the average kinetic/potential energy of a vibrating particle is directly proportional to the square of the amplitude.

Total energy of a vibrating particle:

The total energy is the sum of the kinetic and potential energies of a vibrating particle.

The displacement of a vibrating particle is given by

y = a sin (ωt + α)

sin (ωt + α) = y/a

$$\cos(\omega t + \alpha) = \sqrt{1-\frac{y^2}{a^2}} = \sqrt{\frac{a^2-y^2}{a^2}} = \frac{\sqrt{a^2-y^2}}{a}$$

Velocity, $$v = a\,\omega\cos(\omega t+\alpha) = \frac{a\omega\sqrt{a^2-y^2}}{a} = \omega\sqrt{a^2-y^2}$$

Therefore, the kinetic energy of the particle at the instant the displacement is y,

$$= \frac{mv^2}{2} = \frac{1}{2}\, m\omega^2(a^2-y^2)$$

Potential energy of the vibrating particle is the amount of work done in overcoming the force through a distance y.

Acceleration = - $\omega^2 y$

Force = - $m\omega^2 y$

(The –ve sign shows that the direction of the acceleration and force are opposite to the direction of motion of the vibrating particle.)

$$\text{Therefore}\,P.E = \int_0^y m\omega^2 y.\,dy$$

$$= \frac{m\omega^2 y^2}{2}$$

Total energy of the particle at the instant the displacement is y

$$= K.E + P.E = \frac{1}{2}\, m\,\omega^2(a^2-y^2) + \frac{m\omega^2 y^2}{2}$$

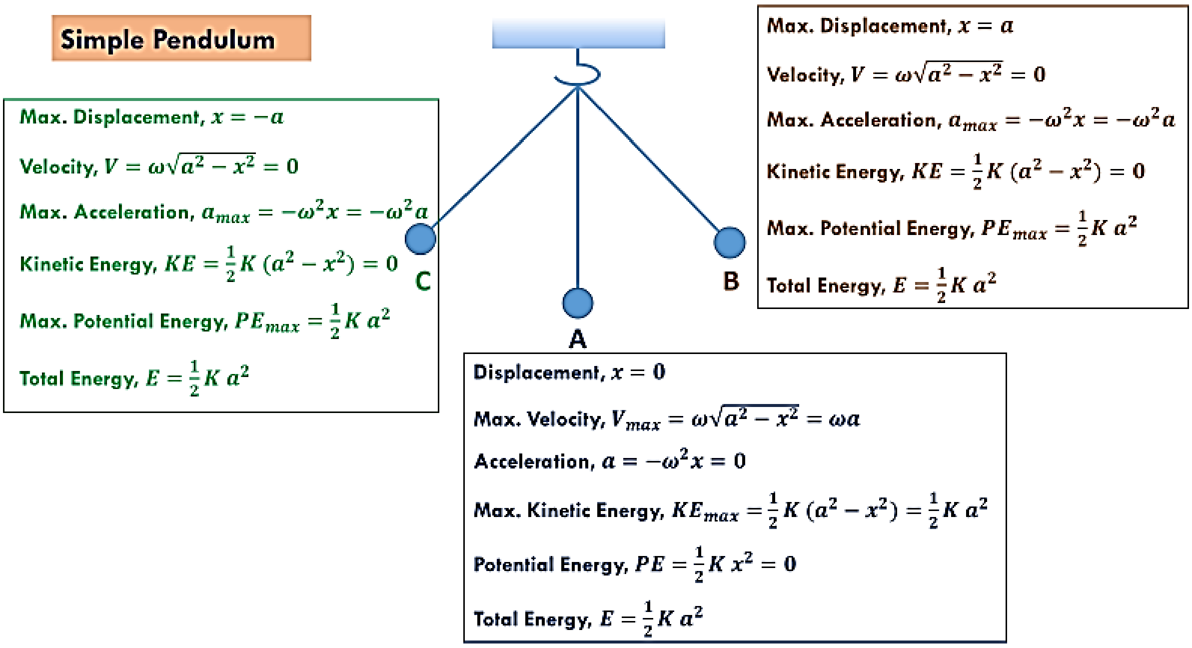
$$= \frac{m\omega^2 a^2}{2}$$

$$= \frac{1}{2} m\,(2\pi n)^2 a^2$$

$$= 2\,\pi^2 m a^2 n^2$$

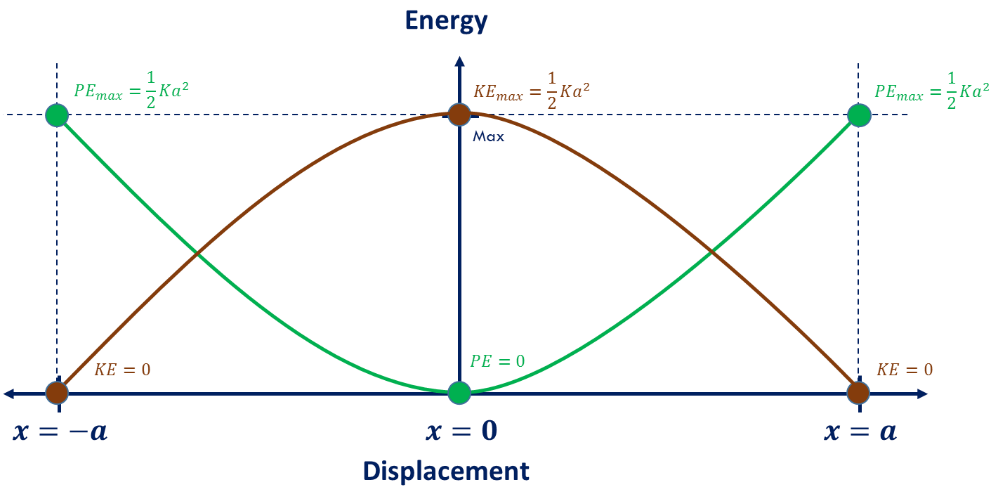
$= \pi^2 m a^2 n^2 = \pi^2 m a^2 n^2$As the average kinetic energy of a vibrating particle, the average potential energy. The total energy of a vibrating particle at any instant is constant.

Investigation of Simple Pendulum:



Graphical Comparison of K.E. and P.E.:

The mechanical oscillator, as it passes through its equilibrium position, its speed and hence its kinetic energy is a maximum. At the maximum displacement, the speed and hence the kinetic energy are both zero. The potential energy will be a maximum when the speed is zero and vice versa. Assuming that there is no friction or air drag, the total energy E of the oscillator must remain constant.



The kinetic energy will be zero at $+a$ and a maximum when $y = 0$, so its graph is an inverted version of the potential energy graph. At any position kinetic + potential energy is a constant E.

Problem-1: *Show that for a particle executing SHM, the instantaneous velocity is* $\omega\sqrt{a^2 - x^2}$ *and instantaneous acceleration is* $-\omega^2 x$.

Solution: For a particle executing SHM,

$$x = a\sin(\omega t + \alpha) \qquad (1)$$

The instantaneous velocity,

$$v = \frac{dx}{dt} = a\omega\cos(\omega t + \alpha) \quad = a\omega\sqrt{1 - \sin^2(\omega t + \alpha)} = \omega\sqrt{a^2 - a^2\sin^2(\omega t + \alpha)}$$

From equation (1)

$$\boldsymbol{v = \omega\sqrt{a^2 - x^2}}$$

The instantaneous acceleration,

$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = -a\omega^2\sin(\omega t + \alpha) == -\omega^2[a\sin(\omega t + \alpha)]$$

$$\boldsymbol{a = -\omega^2 x}$$

Problem-2: *If the acceleration of a particle executing simple harmonic motion is 0.2ms-2 and the amplitude is 0.004m. Find the time period of motion.*

Solution: Given that, $a = 0.2ms^{-2}$

$A = 0.004m$

$T = ?$

We know, $a = \omega^2 A$

$\Leftrightarrow \quad \omega = \sqrt{\frac{a}{A}}$

Again, $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{a}{A}}} = 2\pi\sqrt{\frac{a}{A}} = 2\pi\sqrt{\frac{0.2}{0.004}} = 0.89sec \qquad (Ans.)$

Problem-3: If the maximum velocity of a particle executing simple harmonic motion is 0.2 ms^{-1} and the amplitude is 0.004m. Find the time period of motion.

Solution: Given that, $V_{max} = 0.2ms^{-1}$

$A = 0.004m$

$T = ?$

We know, $V_{max} = \omega A$

$\Leftrightarrow \quad \omega = \frac{V_{max}}{A}$

Again, $T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{V_{max}}{A}} = 2\pi\frac{A}{V_{max}} = 2\pi\frac{0.004}{0.2} = 0.1256\ sec \quad (Ans.)$

Problem-4: If a particle of mass 0.05kg executing simple harmonic motion with frequency 50Hz and if its amplitude is 0.004m. Calculate its average kinetic energy.

Solution: Given that, $m = 0.05ms^{-1}$

$n = 50Hz$

$a = 0.004m$

$KE_{av} = ?$

We know, $KE_{av} = ma^2\pi^2n^2 = 0.05 \times (0.004)^2 \times (3.14)^2 \times (50)^2 = 0.071\,Joule$

$(Ans.)$